IronShield

Stopping fraud before the money moves



Built on IBM LinuxONE

Team Big Iron Brains · IBM Z Datathon 2026 · Real-Time AI for Critical Decisions

Today's fraud is instant. Most detection isn't

A 3-minute scam

Ayesha gets a call from her "bank" and approves a transfer. Within minutes her savings are split across six mule accounts, and they're gone.

Why today's systems miss it

- Batch checks run after the money has moved
- Alerts don't tell analysts why

$579.4B

lost worldwide to fraud scams and bank fraud in 2025

Source: Nasdaq Verafin, 2026 Global Financial Crime Report

$16.6B

cybercrime losses reported to the FBI in 2024, up 33% in a year

Source: FBI IC3 Internet Crime Report 2024

IronShield: four layers of protection, in real time



Real-time scoring

Every transaction gets a fraud-risk score in milliseconds, before the money moves.



Mule-network detection

Graph analysis exposes accounts that collect from many victims and pass money on fast.



Explainable alerts

Plain-English reasons for every flag, so analysts act fast and decisions are auditable.



Privacy-first

Account IDs are hashed before analysis. The model never sees real identities.

Live demo: fraud caught in milliseconds

1 Stream starts

Normal transactions flow onto the dashboard in green.

2 Fraud hits

A suspicious payment turns red within milliseconds.

3 See why

Click it: "amount 12× average, new device, 2 AM".

4 Mule ring

The network view highlights one account fed by many victims.

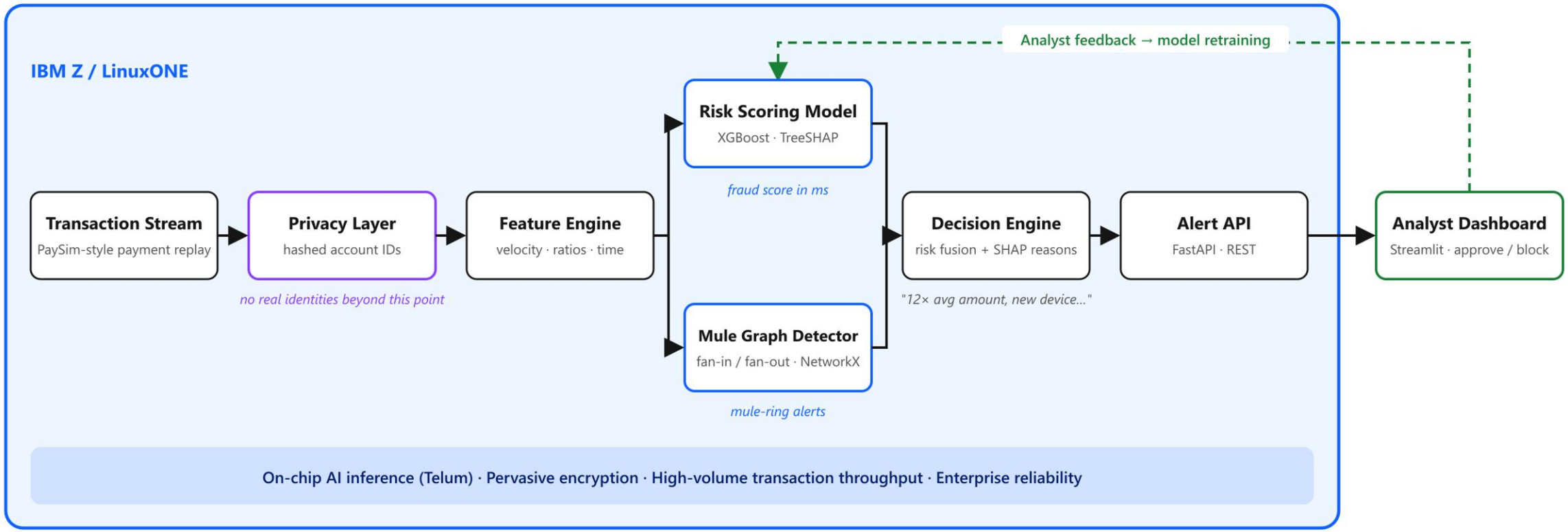
5 Block it

The analyst blocks it, and the decision feeds back into the model.

6 Proof of speed

A live latency counter, running on IBM LinuxONE.

How IronShield works



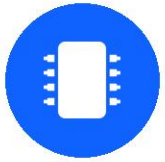
Stream → privacy → features → model + graph → explain → analyst, all running on LinuxONE

Why IBM Z: score fraud where the money moves



Where transactions already live

Score payments in-line, without copying sensitive data to outside systems.



On-chip AI acceleration

IBM Telum processors are built for real-time inference at transaction scale.



Pervasive encryption

Data stays encrypted at rest and in flight, protecting financial records.



Enterprise reliability

Built for the always-on systems banks already trust.

Measured on LinuxONE

0.09 ms

median decision time per payment (p95: 0.43 ms)

7,750 tx/s

throughput on a single core

IBM LinuxONE (s390x), 20,000 payments, full pipeline

Impact: safer payments for everyone

96%

precision: alerts that were real fraud

92%

recall: share of fraud caught

19/19

mule rings caught

Held-out test on IBM LinuxONE: 27,066 future transactions (342 frauds), synthetic PaySim-style data, time-ordered split

Who benefits

Banks & fintechs

Fewer losses and faster investigations

Mobile-money providers

Protect fast-growing digital payments

Everyday users

Especially the newly banked, who are hit hardest

Thank you

IronShield: Big Iron, with a brain.

What's next

- Connect to real payment rails
- Federated learning across banks
- IBM Z on-chip inference in production

GitHub: github.com/huzaifaguru/ironshield
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